ME 5309 – Mid-Semester Examination: ANSWER SHEET

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Questions attempted: Q1, Q2, Q3, Q4 (Q1 struck out – not to be considered)

Question 1 [CROSSED OUT BY STUDENT]

~~*Given: sensor 1280 px wide, pixel pitch $p = 5\ \mu m$ (square), lens $f = 16$ mm, standoff $Z = 400$ mm; force sensor 10 mV/N, 12-bit ADC over 0–5 V; plant $\dot{x} = -2x + u$ with zero-order hold, $T = 0.1$ s.*~~

~~**(a)** (i) $f_x = f/p = 16\times10^{-3} / 5\times10^{-6} =$ **3200 px**.~~

~~(ii) Sensor width = 1280 × 5 µm = 6.4 mm. Using similar triangles, FOV = $W_s \cdot f / Z = 6.4 \times 16/400 =$ **0.256 mm**.~~

~~Distance per pixel = FOV/1280 = 0.256/1280 = **2×10^{-4} mm** = 0.2 µm.~~

~~**(b)** LSB = 5/4096 = 1.2207 mV.~~

~~ΔF = LSB / S = 1.2207 mV / 10 mV/N = **0.0122 N**.~~

~~**(c)** With u held constant over [kT, (k+1)T] (zero-order hold) the variation-of-constants formula gives $x((k+1)T) = e^{AT}x(kT) + \int_0^T e^{A(T-s)} B\, u(kT)\, ds$. Since u(kT) is constant it leaves the integral, so $F = e^{AT}$ and $G = \int_0^T e^{A\sigma} B\, d\sigma = A^{-1}(e^{AT} - I)B$ for invertible A.~~

~~Substitutting $A = -2$, $B = 1$, $T = 0.1$ s: $F = e^{-0.2} = 0.81873$. $G = (0.81873 - 1)/(-2) = 0.090635$.~~

~~Sanity check: the DC gain of the continuous plant is $-A^{-1}B = 0.5$, and the discrete DC gain $G/(1 - F)$ = 0.090635/0.18127 = 0.5 ✓.~~

~~**$x_{k+1} = 0.8187\, x_k + 0.0906\, u_k$**~~

~~*Sanity chek: a 160 mm field of view across 1280 pixels is plausible for a 400 mm standoff and a 16 mm lens.*~~

~~*(Cancelled - please do not evaluate this question.)*~~

Question 2

Given: $J = 0.02$ kg·m², $b = 0.05$ N·m·s/rad, $m = 1$ kg, $\ell = 0.25$ m, $g = 9.81$ m/s². So $mg\ell = 1 \times 9.81 \times 0.25 = 2.4525$ N·m (used throughout).

(a) The model is second order, so I choose the position and velocity as states: $x = [x_1, x_2]^T = [\theta, \dot{\theta}]^T$, with input τ.

Rewriting $J\ddot{\theta} + b\dot{\theta} + mg\ell \sin\theta = \tau$ gives the nonlinear state-space model: $\dot{x}_1 = x_2$, $\dot{x}_2 = (1/J)(\tau - b\, x_2 - mg\ell \sin x_1)$, i.e. $\dot{x} = f(x, \tau)$.

Equilibria for constant τ_0 need $\dot{x} = 0$: the first equation gives $x_2 = 0$, and the second then gives $mg\ell \sin\theta_e = \tau_0$, i.e. $\sin\theta_e = \tau_0/(mg\ell)$.

This has a solution only if $|\tau_0/(mg\ell)| \le 1$, so **$|\tau_0| \le mg\ell = 2.4525$ N·m**. For $|\tau_0| < mg\ell$ there are two equilibria per revolution (θ_e and $\pi - \theta_e$); at $|\tau_0| = mg\ell$ they merge, and beyond it the joint cannot be held statically.

(b) Linearising: $A = [0, 1;\ \partial f_2/\partial x_1,\ \partial f_2/\partial x_2]$.

$\partial f_2/\partial x_1 = -(mg\ell/J)\cos\theta_e = -122.625 \times 0.8660 = -106.2$. $\partial f_2/\partial x_2 = -b/J = -2.5$.

$A = [0, 1; -106.2, -2.5]$, $B = [0; 1/J] =$ **$[0; 50]$**.

(c) Set all time derivatives to zero in the closed loop. With $\tau = K_p(\theta_d - \theta) - K_d\dot{\theta}$ and $\dot{\theta} = \ddot{\theta} = 0$ the damper term vanishes (so K_d has no effect on the final position) and the dynamics reduce to $mg\ell \sin\theta_{ss} = K_p(\theta_d - \theta_{ss})$.

Why $\theta_{ss} \neq \theta_d$: if the joint were at rest exactly at $\theta_d = 90°$ the PD torque would be $K_p \cdot 0 - K_d \cdot 0 = 0$, yet gravity pulls with $mg\ell \sin 90° = 2.4525$ N·m. The net torqe would be non-zero, so the joint could

not be at rest. A purely proportional-derivative law can only hold a load if there is a steady error.

Let $e = \theta_d - \theta_{ss}$ (in rad). Then $\sin\theta_{ss} = \sin(\pi/2 - e) = \cos e$, and the equation becomes $50\,e = 2.4525\cos e$. Fixed-point iteration: $e_0 = 2.4525/50 = 0.04905$ (taking $\cos e \approx 1$); $e_1 = 0.04905 \times \cos(0.04905) = 0.04899$; $e_2 = 0.04905 \times \cos(0.04899) = 0.04899$, converged.

$e = 0.04899$ rad $\times (180/\pi) = 2.807°$, so $\theta_{ss} = 87.19°$. **Steady-state error ≈ 2.81° (the arm sags below the target).** This is small because K_p is much larger than $mg\ell$ ($K_p/mg\ell \approx 20$).

(d) Stedy state: $\dot{\theta} = \ddot{\theta} = 0$ and the control law gives $\tau = K_p(\theta_d - \theta_{ss}) + mg\ell\sin\theta_{ss}$. Plugging into the plant, $mg\ell\sin\theta_{ss} = K_p(\theta_d - \theta_{ss}) + mg\ell\sin\theta_{ss}$, so $K_p(\theta_d - \theta_{ss}) = 0$ and **$\theta_{ss} = \theta_d$: steady-state error is zero**.

Poles: linearise the closed loop about θ_d. Before compensaton, $J\ddot{\theta} + (b + K_d)\dot{\theta} + (K_p + mg\ell\cos\theta_d)\delta\theta = 0$, characterstic polynomial $J\lambda^2 + (b + K_d)\lambda + K_p + mg\ell\cos\theta_d$. After compensation the added term $mg\ell\sin\theta$ cancels the plant's gravity term exactly, leaving $J\lambda^2 + (b + K_d)\lambda + K_p$.

The two differ only by $mg\ell\cos\theta_d$, and for $\theta_d = 90°$ this is $2.4525 \times 0 = 0$. So **the closed-loop poles are unchanged** at this operating angle. I should stress this is because $\cos 90° = 0$; if θ_d were, say, 60°, the compensation would raise the stiffness by 61.3×0.02... i.e. it would move the poles.

Question 3

Given: $A = [0, 1; 0, -2]$, $B = [0; 1]$, $C = [1, 0]$, $f_s = 1$ kHz so the sample time is $T = 1/f_s = 1$ ms.

(a) Reason 1 – noice amplification: the backward difference $\hat{\omega}_k = (\theta_k - \theta_{k-1})/T$ has frequency response $(1 - e^{-j\omega T})/T$, whose magnitude grows with frequency ($\approx \omega$ for small ωT), i.e. it is a high-pass operation. Any noise on the encoder signal therefore appears in the velocity multiplied by roughly $1/T = 1000\ s^{-1}$, and can swamp the true velocity.

Reason 2 – quantisation: the encoder only reports integer counts, so $\Delta\theta$ is an integer multiple of the count size q. Dividing by T = 1 ms gives velocity steps of q/T; for a 4096-count encoder $q = 2\pi/4096 = 1.53\times10^{-3}$ rad, so one step is 1.53 rad/s. At low speed the estimate flips between 0 and one step (a noisy staircase), exactly where the damping term of the PD law is needed.

(A further, smaller issue is the half-sample delay of a backward difference, but the two reasons above are the main ones.)

(b) The analogue prototype is the first-order low-pass $H(s) = \omega_c/(s + \omega_c)$ with $\omega_c = 2\pi f_c = 2\pi \times 20 = 125.66$ rad/s. The bilinear (Tustin) transform replaces s by $(2/T)(z - 1)/(z + 1)$ with $T = 1/f_s = 0.001$ s.

Define $a = \omega_c T/2 = 125.66 \times 0.001/2 = 0.062832$. Then $H(z) = \omega_c / [(2/T)(z-1)/(z+1) + \omega_c] = a(z + 1) / [(1 + a)\,z - (1 - a)] = a(1 + z^{-1}) / [(1 + a) - (1 - a)\,z^{-1}]$.

Dividing through by $(1 + a)$: $a/(1 + a) = 0.062832/1.062832 = 0.059117$ and $(1 - a)/(1 + a) = 0.937168/1.062832 = 0.881767$. Hence with u = differenced signal and y = filtered velocity: **$y_k = 0.05912\,(u_k + u_{k-1}) + 0.88177\,y_{k-1}$**. (Frequency pre-warping is neglected since $f_c \ll f_s/2$.)

Verification of the DC gain: set u and y constant ($z = 1$): $y = 2(0.05912)u + 0.88177\,y \Rightarrow y(1 - 0.88177) = 0.11824\,u \Rightarrow y/u = 0.11824/0.11823 = 1.000$. In general $H(1) = 2a/[(1 + a) - (1 - a)] = 2a/2a = 1$ exactly, so the steady-state output equals the commanded input ✓.

(c) First check observability: $O = [C; CA] = [1, 0; 0, 1]$, full rank, so the observer poles can be placed arbitrarily.

With $L = [\ell_1, \ell_2]^T$ and $C = [1, 0]$, $LC = [\ell_1, 0; \ell_2, 0]$, so $A - LC = [-\ell_1, 1; -\ell_2, -2]$.

Characteristic polynomal: $\det(sI - (A - LC)) = \det[s + \ell_1, -1; \ell_2, s + 2] = (s + \ell_1)(s + 2) - (-1)(\ell_2) = s^2 + (2 + \ell_1)\,s + (2\ell_1 + \ell_2)$.

Both estimator poles at $s = -10$ means the desired polynomal is $(s + 10)^2 = s^2 + 20\,s + 100$. Comparing coefficients: s^1: $2 + \ell_1 = 20 \Rightarrow \ell_1 = 18$. s^0: $2\ell_1 + \ell_2 = 100 \Rightarrow \ell_2 = 100 - 36 = 64$.

L = [18, 64]^T. Verification: $A - LC = [-18, 1; -64, -2]$ has trace -20 and determinant $(-18)(-2) - (1)(-64) = 36 + 64 = 100$, so the polynomial is $s^2 + 20s + 100 = (s + 10)^2$ ✓.

(d) Advantage of the observer: it uses the model and the input u, so there is no lag in the estimat, which a filter always introduces.

The filter is safer when the sampling rate is very high, because the obsever would not be able to run fast enough.

Question 4

Given: plant $\theta^{\cdot\cdot} + a\theta^{\cdot} + c\theta = b\,u$ with $a = 2$, $c = 6$, $b = 1$; controller $u = N_{ff}\,r - k_1\theta - k_2\theta^{\cdot}$.

(a) Substitute the control law into the plant: $\theta^{\cdot\cdot} + a\theta^{\cdot} + c\theta = b\,N_{ff}\,r - b\,k_1\,\theta - b\,k_2\,\theta^{\cdot}$. Moving the feedback terms to the left: $\theta^{\cdot\cdot} + (a + b\,k_2)\,\theta^{\cdot} + (c + b\,k_1)\,\theta = b\,N_{ff}\,r$.

Numerically, $a + b\,k_2 = 2 + 2 = 4$ and $c + b\,k_1 = 6 + 2 = 8$, so the characteristic equation is $s^2 + 4s + 8 = 0$. Roots: $s = [-4 \pm \sqrt{(16 - 32)}]/2 = -2 \pm 2j$. They have negative real parts, so the loop is stable; $\omega_n = \sqrt{8} = 2.83$ rad/s and $\zeta = 4/(2 \times 2.83) = 0.707$.

Transfer function from r to θ: $T(s) = b\,N_{ff} / (s^2 + 4s + 8)$. DC gain ($s = 0$): $T(0) = N_{ff}/8$. For unitiy DC gain, **$N_{ff} = 8$**. (Note that this is larger than 1 because the gravity-related stiffness c and the feedback k_1 both resist the displacement.)

(b) With the actual $b = 0.75$ and unchanged $N_{ff} = 8$, $k_1 = k_2 = 2$, the closed loop becomes $\theta^{\cdot\cdot} + (a + bk_2)\theta^{\cdot} + (c + bk_1)\theta = bN_{ff}\,r$, i.e. $\theta^{\cdot\cdot} + 3.5\theta^{\cdot} + 7.5\theta = 6\,r$.

For a unit step the stedy state is found by setting derivatives to zero: $7.5\,\theta_{ss} = 6 \times 1$, so $\theta_{ss} = 6/7.5 =$ **0.8**.

Tracking error $= (r - \theta_{ss})/r \times 100\% = (1 - 0.8) \times 100 =$ **20%**. A common mistake would be to say the output falls by the same 25% as the actuator gain; that is wrong here, because b also appears in the feedback stiffness $c + bk_1$, which decreases from 8 to 7.5 and partially compensates. The poles also change: $s^2 + 3.5s + 7.5 \rightarrow s = -1.75 \pm 2.11j$, still stable.

(c) Substituting the control law and differentiating once: $s^3 + (c + bk_2)s^2 + (a + bk_1)s + bK_I$.

Desiered: $(s + 4)(s^2 + 4s + 8) = s^3 + 8s^2 + 24s + 32$.

$6 + k_2 = 8 \Rightarrow$ **$k_2 = 2$**; $2 + k_1 = 24 \Rightarrow$ **$k_1 = 22$**; $K_I =$ **32**.

(d) Stedy state: $\xi^{\cdot} = 0$, so $r = \theta_{ss}$; the parameters b and c do not enter, hence zero error for any b and c as long as it is stabel.

Failure mode: windup when the actuator saturates. Remedy: reduce K_I to zero (or use a smaller gain) while saturated.

Remark: with integral action the exact value of N_{ff} matters less because the integrater enforces tracking.